required for workers is expressed in hours. Such measures are not suitable for comparing the variability of the two data sets which are expressed in different units of measurements.

6.3.1 Range

Raw Data:

Range is defined as difference between the largest and smallest observations in the data set. Range(R) = Largest value in the data set (L) –Smallest value in the data set(S)

$$R = L - S$$

Grouped Data:

For grouped frequency distribution of values in the data set, the range is the difference between the upper class limit of the last class interval and the lower class limit of first class interval.

Coefficient of Range

The relative measure of range is called the coefficient of range

Co efficient of Range = (L- S) / (L+ S)

Example 6.2

The following data relates to the heights of 10 students (in cms) in a school. Calculate the range and coefficient of range.

158, 164, 168, 170, 142, 160, 154, 174, 159, 146

Solution:

L=174 S=142

Range = L – S = 174 – 142 = 32

Coefficient of range = (L - S)/(L + S)

= (174-142) / (174+142) = 32 / 316 = 0.101

Example 6.3

Calculate the range and the co-efficient of range for the marks obtained by 100 students in a school.

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